1901.2a

- IEEE 1901.2a-2013 is a wired technology that updates the original IEEE 1901.2 standard. This is the Narrow-band Power Line Communication standard (NB-PLC).
- Both continuous and direct current electric power lines are specified in IEEE 1901.2a.
- The data rate may be increased to 500 kbps.

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- **NB-PLC is frequently used in the following scenarios**

- **Smart metering –**
  The NB-PLC can automate the reading of utility meters such as electric, gas, and water meters.
- **Distribution automation –**
  The NB-PLC may be used for distribution automation, which helps monitoring and controlling all the power grid's components.
- **Public lighting –**
  It is a frequent application for NB-PLC, which includes lights seen in cities and along roadways, highways, and public spaces like parks.
- **Electric vehicle charging stations –**
  NB-PLC may be used to power charging stations for electric vehicles.
- **Micro-grids –**
  NB-PLC may be used to create micro grids, which are small energy grids that can run independently of the main grid.
- **Renewable energy –**
  The NB-PLC may be utilized in solar, wind, hydropower, and geothermal heat applications. There are several PLC standards, but the lack of a low-frequency PLC solution prompted the creation of IEEE 1901.2a. Below 500 kHz.
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Features

Standardization and Alliances

- Poor dependability, limited throughput, lack of management, and poor compatibility affected the earliest generations of NB-PLC systems.
- Orthogonal frequency-division multiplexing is used in the current NB-PLC specifications (OFDM). Digital data is encoded using OFDM on multiple carrier frequencies.

Physical Layer

- NB-PLC is defined for frequency bands from 3 to 500 kHz.
- IEEE 1901.2a supports the largest set of coding and enables both robustness and throughput.
- Tone maps and modulations for all bands, such as robust modulation (ROBO), differential binary phase-shift keying (DBPSK), differential quadrature phase-shift keying (DQPSK), differential 8-point phase shift keying (D8PSK), and optionally 16 quadrature amplitude modulation (16QAM) for some bands, are included in the standard.

MAC Layer

- The IEEE 1901.2a MAC frame format is related to the IEEE 802.15.4 MAC frame.
- Information elements are one of the important components that have been moved over from IEEE 802.15.4e to IEEE 1901.2a.
- Segment Control is a field in IEEE 1901.2. This deals with the segmentation or fragmentation of upper-layer packets that are bigger than the MAC protocol data unit can carry (MPDU).

3	2	1	0/2	0/2/8	0/2	0/2/8	0/5/6 10/4	variable	variable	2(By
Segment control	Frame control	Seq. number	Destination PAN Identifier	Destination. Address	Source PAN Identifier	Source Address	Auxiliary Security Header	Information Elements	Frame Payload	Frame check sequence

MAC Header(MHR)	MAC Protocol Data unit(MPDU) payload	MAC Footer (MFR)

MAC frame format for IEEE1901.2a

Topology for IEEE 1901.2a :

- IEEE 1901.2a scenarios and deployment topologies are based on physical power lines. S
- ignal transmission is limited by variables such as noise, interference, distortion, and attenuation, just as it is with wireless technology. Because these variables grow more prominent as distance increases, most NB-PLC implementations have a mesh topology.
- Mesh networks provide the benefit of allowing devices to relay traffic from other devices, allowing greater distances to be separated.
- The IEEE 1901.2a standard allows any upper-layer protocol to be used. As a result, IPv6 6LoWPAN and RPL (routing protocol) IPv6 variants are supported. These protocols allow mesh networks to be created over PLC using network layer routing.

Constrained Nodes and Networks

A constrained network exhibits below characteristics:

- Low bit-rate/throughput
- High packet loss and high variability of packet loss
- Highly asymmetric link characteristics
- Lack of advanced network services like multi-cast

The possible reasons for the characteristics of constrained network are below:

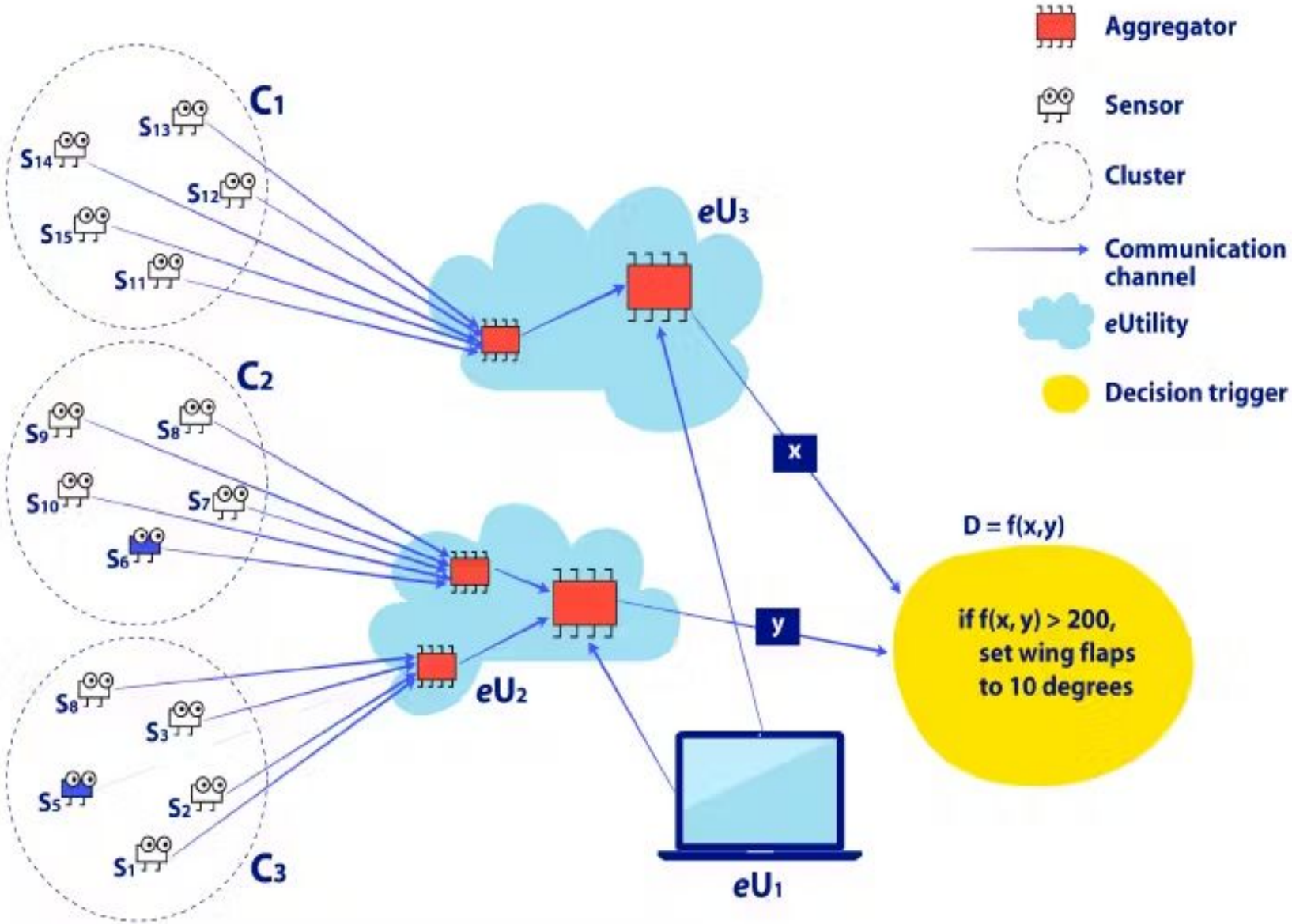
- Cost
- Constraints posed by the nodes
- Physical constraints, e.g., power constraints or environmental constraints
- i-Technology constraints such as using legacy technologies (outdated hardwares)

Constrained node network

- A constrained network is composed of a significant portion of constrained nodes. Mostly, these **constrained node networks are deployed in the edge network of an IoT system.**

Building Blocks of Constrained Node Network

- **Sensors** – A sensor is an electronic utility that measures physical properties and gives electrical output. E.g. temperature sensor and acceleration sensor.
- **Actuators** - An actuator is a device which takes electronic input and gives physical output; e.g. motors.
- **Cluster** - A cluster is a group of sensors and actuators.
- **Communication Channel** - A communication channel is a medium through which data is transferred; e.g. wired or wireless.
- **Aggregators** – An aggregator is a device used to aggregate all the data from sensors and sometimes give the command to actuators. **It is the gateway device.**
- **eUtility** – It is a software or hardware or services which support aggregators for feeding data and computing.
- **Decision Trigger** – A decision trigger is a software which performs computing and takes action if needed.



Building blocks of constrained node network

LORAWAN (Long Range Wide Area Network)

- LoRaWAN uses an unlicensed frequency band for long range communication.
- It can be used upto 3 miles and upto 10 miles in line of sight.
- Range can be extended in LoRaWAN using LoRaX.
- Since LoRaWAN devices operate at low power and low bandwidth the battery can run upto 10 years or more.

Categories of LoRaWAN

LORAWAN Class A devices

- It has the longest battery life.
- It stays in sleep mode majority of the time
- It can receive messages only if it has sent messages.
- Example: fire alarm, flood detectors etc.

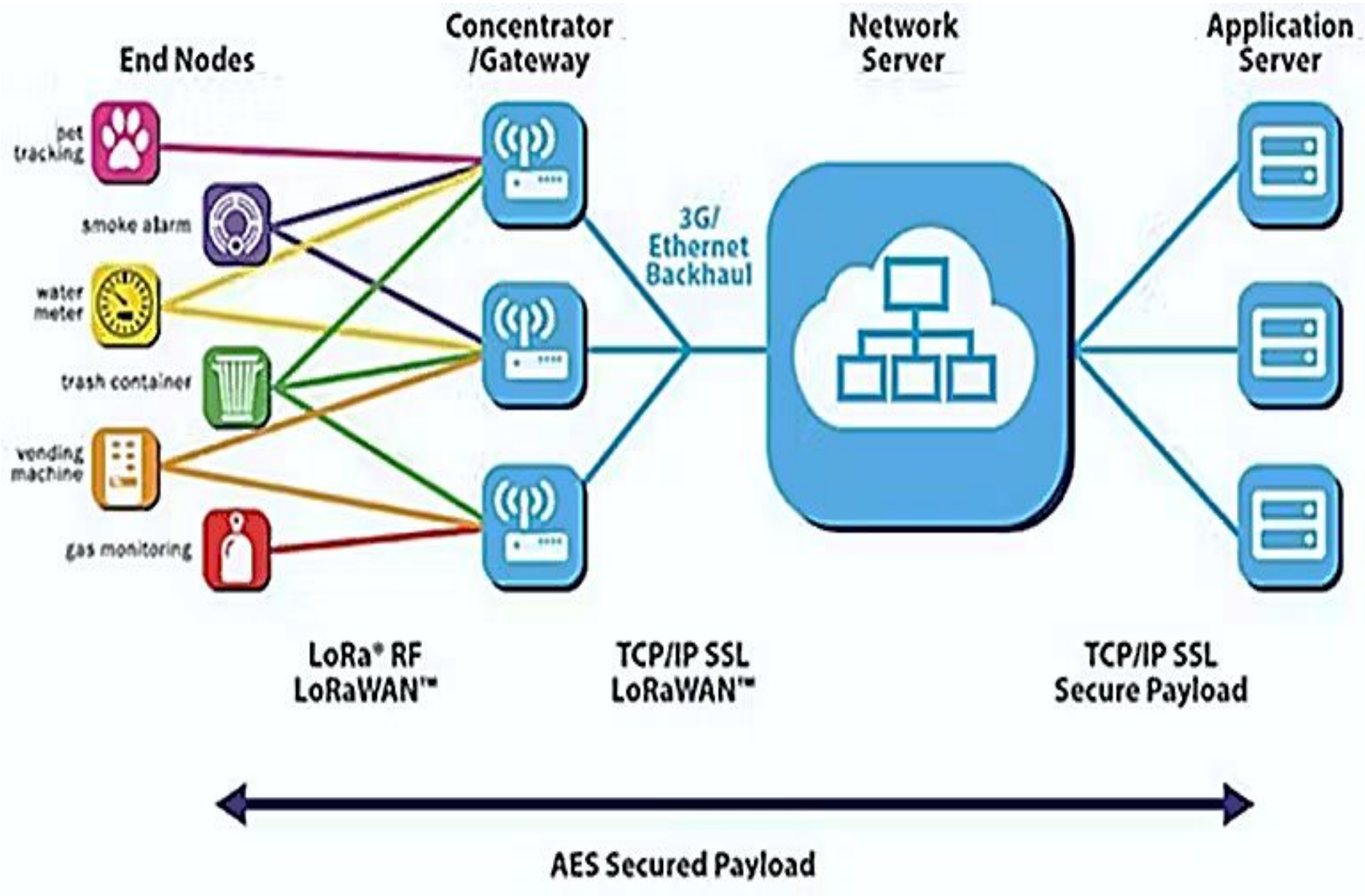
LORAWAN Class B devices

- It has the average battery life.
- It listens to the network periodically
- It can receive messages even if it has not sent messages.
- Example: metering temperature, humidity etc.

LORAWAN Class C devices

- It has low battery life.
- It listens to the network continuously.
- It has minimum latency for messages.
- Example: traffic monitoring, real time applications etc.

Network Server



- **The LoRa network has four elements:**
- **End-nodes** are elements of the LoRa ecosystem that gather sensor data and transmit/receive the data. They're generally remotely connected and are battery-powered.
- **The gateway** is a transparent bridge between the end-nodes and network server. Typically, end-nodes use LoRaWAN to connect to the gateway, while the gateway uses high-bandwidth networks such as Wi-Fi, Ethernet, or cellular to connect to the networks.
- **A network server** connects to multiple gateways. It gathers data from the gateways and filters out duplicate messages, decides which gateway should respond to end-node messages, and adapts data rates to extend battery life of end-nodes.
- **The application server** collects data from end-nodes and controls the actions of the end-node devices.

Common applications of LoRaWAN

- Street Lighting
- Smart Parking
- Waste management
- Water level and flood management
- Smart Public Transport
- Air Quality Control

Advantages

- High density network
- Low power consumption
- Higher data rates (~250kbps)
- Low cost
- Low bandwidth
- Extended coverage
- High mobility

RPL Protocol (Routing over low power lossy network)

- RPL stands for Routing Protocol for Low Power and Lossy Networks for heterogeneous traffic networks.
- It is a routing protocol for Wireless Networks.
- This protocol is based on the same standard as by Zigbee and 6 Lowpan is IEEE 802.15.4 It holds both many-to-one and one-to-one communication.
- Lossy network in RPL means a high packet drop rate.
- RPL is mainly used with IPV6 but it can also be used with IPV4.
- RPL is a distance vector routing protocol, it is based on DODAG (distance oriented directed acyclic graphs)
- A **distance vector routing protocol** is a method that routers use to determine the best path to send data by measuring the distance and direction to a destination.
- DODAG is a tree structure that organizes the network topology.

RPL Terminology

Directed Acyclic Graphs

- It is a graph that contains no cycle, Such graphs can be seen in Spanning Trees.

Root

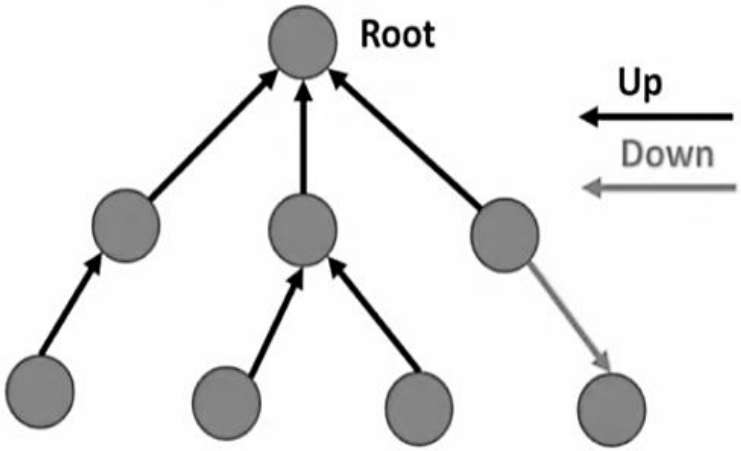
- It is a destination node of DAG. It has no outgoing edge.

Up

- It is any edge that is directed towards the Root.

Down

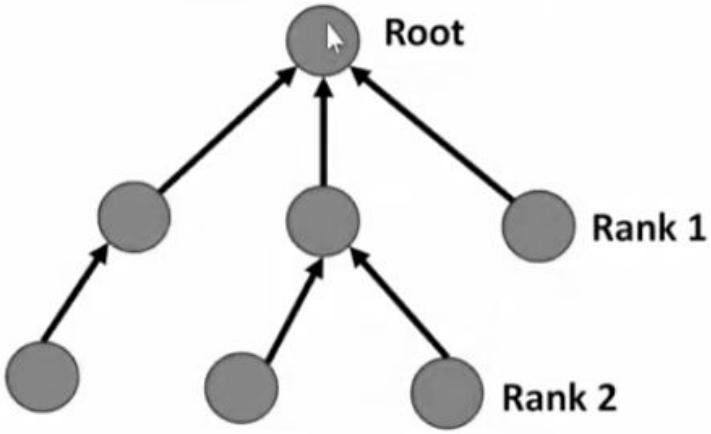
- It is any edge that is directed away from the Root.



RPL Terminology

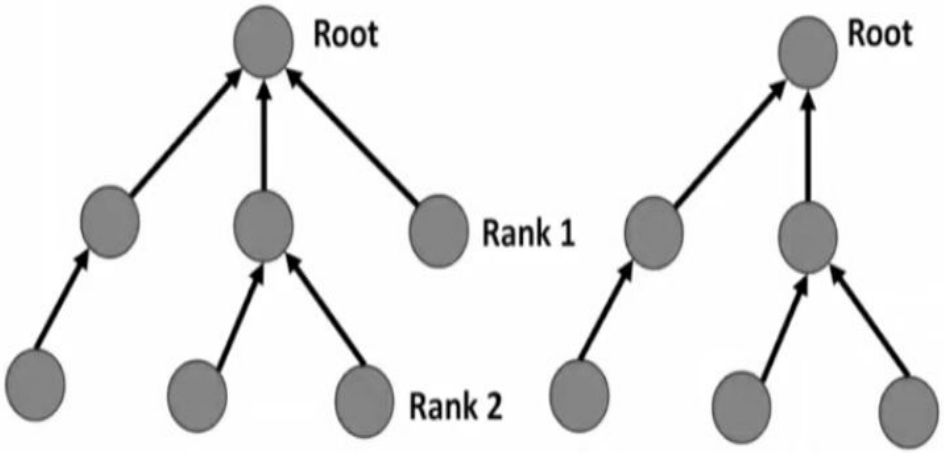
- Destination Oriented Directed Acyclic Graphs (DODAG)
  - This is a special kind of DAG where each node wants to reach a single destination.
- Objective Function
  - It helps us to decide whether we are near to the root or away from it.
  - It is decided by the programmer or designer. It is something that we want to minimize. It can be a latency of transmission, distance of transmission, number of hopes.

- Rank
  - It is number of hopes from Root.



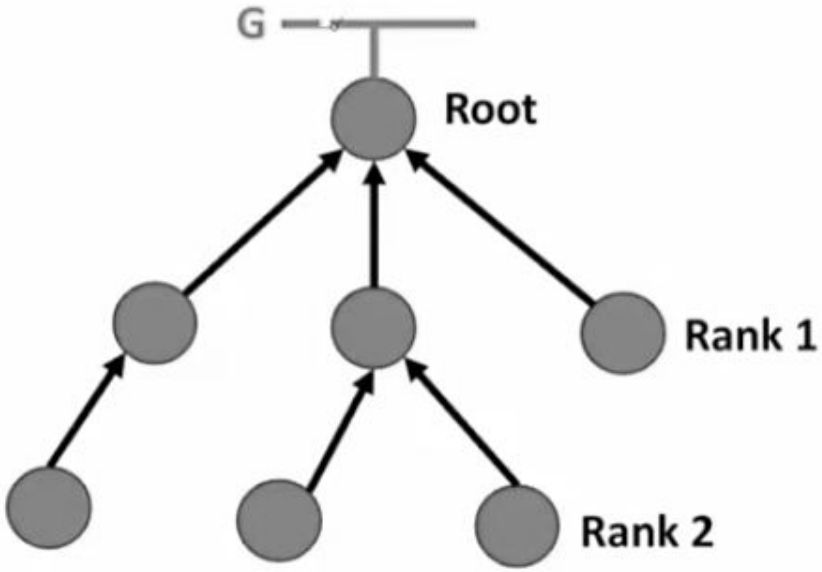
RPL Terminology

- RPL Instance
  - When we have one or more DODAGs, then each DODAG is an instance.
- DODAG ID
  - Each DODAG has IPv6 ID of 128 bits.
  - This ID is given to its root only and it stays until the route doesn't change.
- DODAG Version
  - Each new shape of DODAG means a new version.

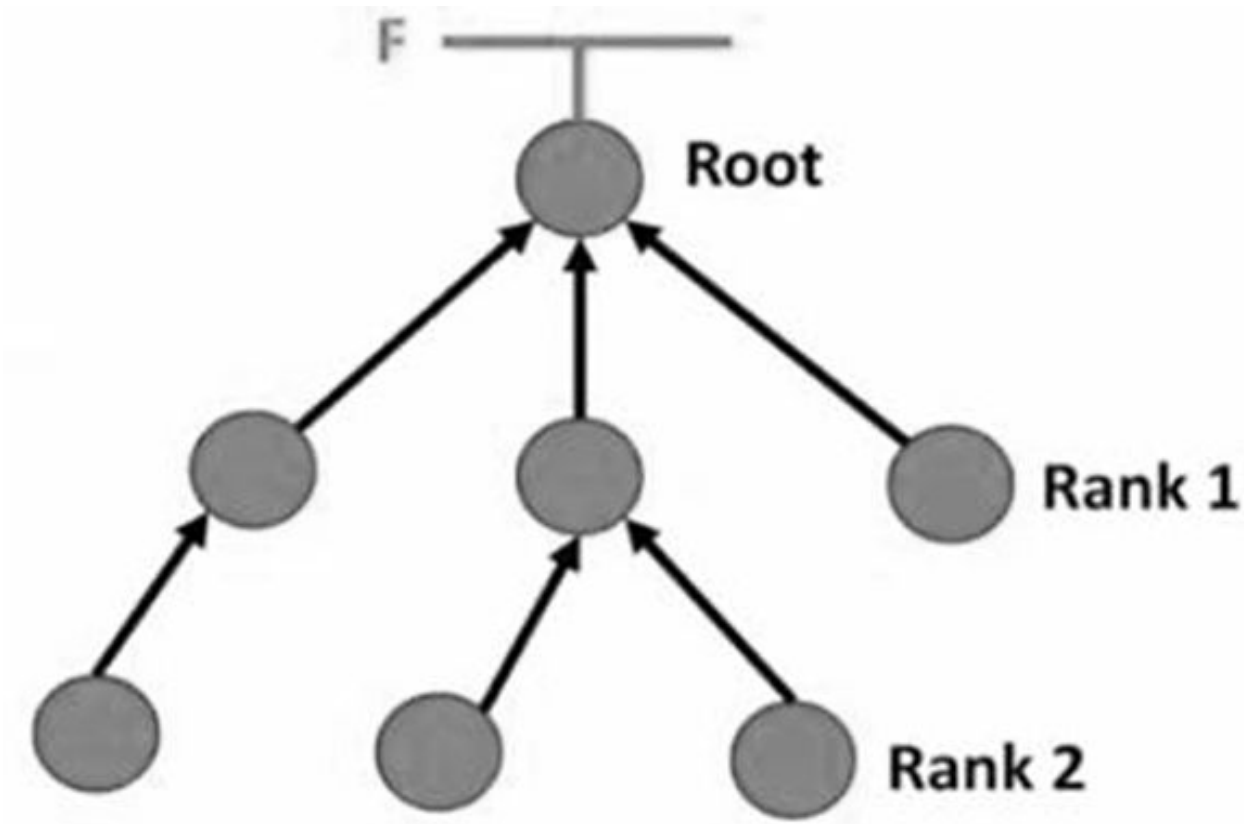


RPL Terminology

- RPL Goal
  - It is to provide the best route path using DODAG.
  - The aim is to minimize the objective function.
- Grounded
  - When DODAG reaches its goal, it is known as grounded.
  - G in the figure shows grounded DODAG.

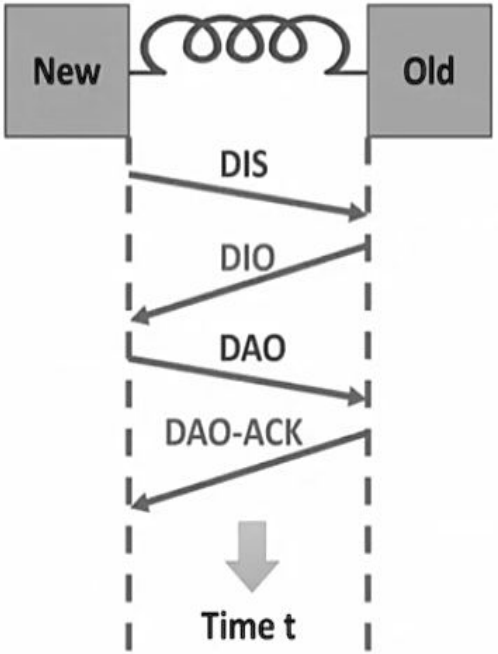


Floating node is the node which has not yet grounded. When DODAG is not connected or it has not reach its goal it is called as floating node.

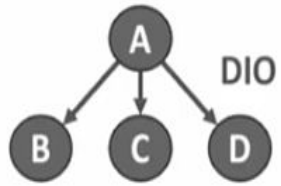
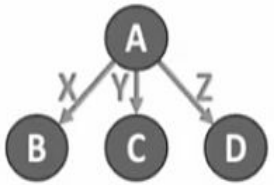
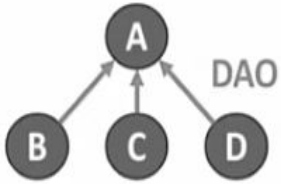
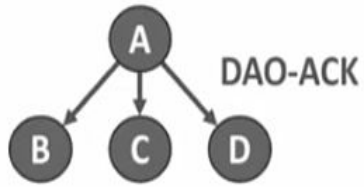
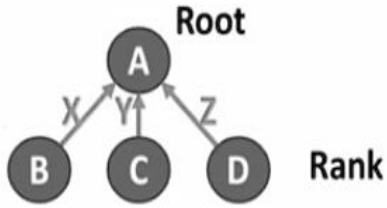


RPL Control Messages

- DODAG Information Solicitation (DIS)
  - When no announcement is heard, and if a node wants to join DODAG, it sends a control message, for that it wants to know if any DODAG exists.
- DODAG Information Object (DIO)
  - This message is multi-casted downwards. Node in DODAG may multicast this message, which lets other nodes know about it, things like whether the node is grounded or not, whether it storing or Non-storing, and it announces other nodes "if they are interested to join, please let me know."
- DODAG Advertisement Object (DAO)
  - It is a request sent by the child node to the root node. This message requests to allow the child node to join to a DODAG.
- DAO-ACK
  - It is a response sent by a root or parent to the child node. This response can either be a YES or NO.



- Consistency Check (CC)
  - It deals with security.
  - It gives consistency to given DODAG.

DODAG Formation

- Root in a DODAG is a special node. All nodes don't have the capability to be a root node. A root node is programmed for it.
- Suppose we have 4 nodes: A, B, C, and D. To form DODAG, following steps are taken.
- Node A (Root Node) Multicasts DIOs.
- Upon receiving DIOs by rest nodes (B, C & D), rest nodes will be having idea of distances with respect to node A.
- Then rest nodes (B, C & D) send DAOs to Root Node A.
- Root Node A accepts DAOs and it gives DAO-ACK to rest nodes.
- Another similar round will be initiated by the nearest to the root. This process continues till the identification of the shortest distance in the given network.